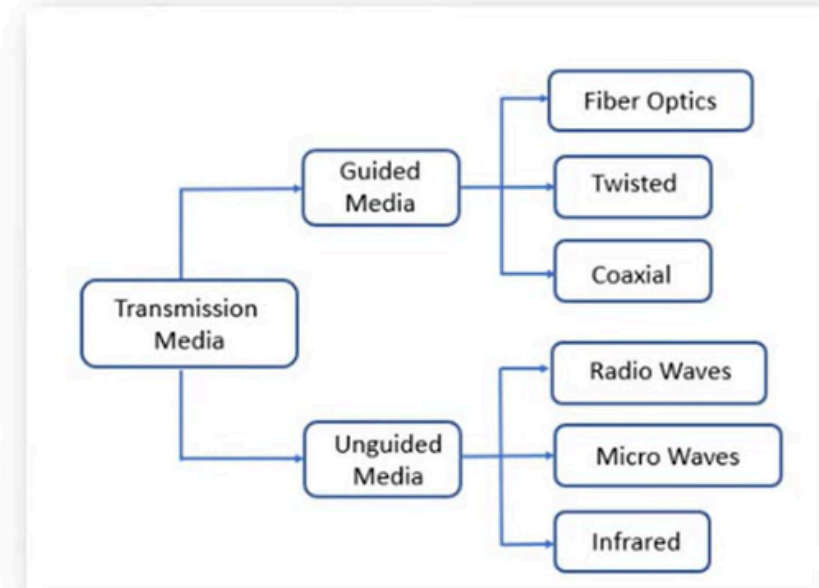


About Transmission Medias

- It refer to physical pathways through which data is transmitted from one device to another.
- These pathways can be wired or wireless.
- The choice of medium depends on factors like distance & speed etc.



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1.1 Twisted Pair Cable

- Here, pair of copper cables twisted with each other.
- It has 2 separately insulated conductor wires wound about each other.

Type 1: Unshielded Twisted Pair

- Mostly used in Telecommunication.
- Low cost, simple to install & high speed.
- It is categorized between 4 to 200 mbps speed.

Type 2: Shielded Twisted Pair

- Mostly used in Ethernet, Industrial control systems, Medical equipment.
- Higher data transmission rate then UTP.
- Installation is easy & Expensive then UTP.



1.2 Coaxial Cable

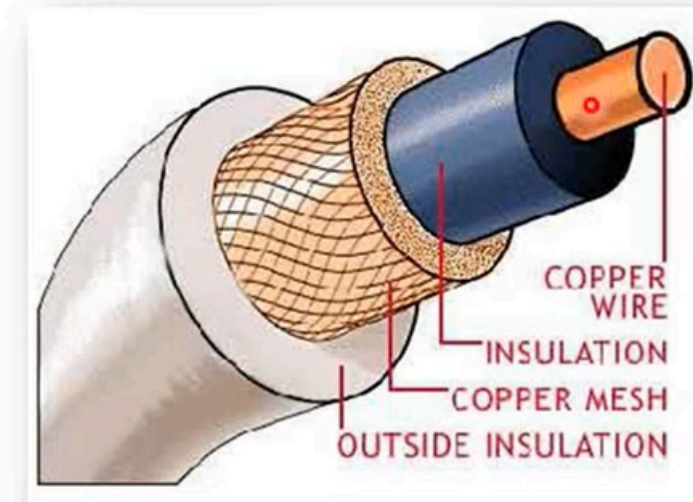
- The inner conductor made up of copper & the outer conductor is made up of copper mesh.
- The middle core is made up of non-conductive cover that separates the inner conductor from the outer conductor.
- It has a higher frequency as compared to Twisted pair cable.
- Easy to install but Expensive.

Type 1: Baseband transmission

- It's transmitting a single signal at high speed.
- Mainly used in LAN.

Type 2: Broadband transmission

- It's transmitting multiple signals simultaneously at high speed.
- Mainly used in Modem & TV systems.

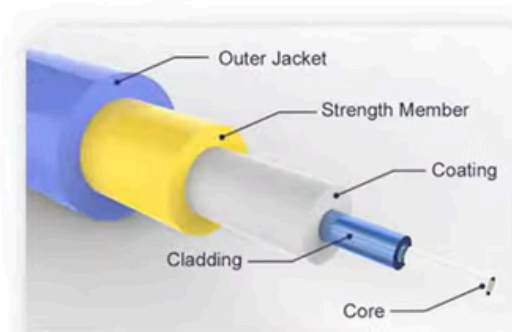


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1.3 Fiber Optic Cable

- It used to send the data by pulses of light.
- It performs well in long-distance data transmission and telecommunications.
- They can carry a large amount of information and transmit data at a high speed.
- It is high cost & difficult for installation & maintenance.

- ✓ **Core** is designed to transport light.
- ✓ **Cladding** is a thin layer that helps transmit data through the fiber.
- ✓ **Coating** gives further protection for the fiber core.
- ✓ **Strength Member** keep fiber core from damage from installation.
- ✓ **Outer Jacket** is designed to protect cable from environmental damage.



➤ **Application:** Internet, Telephone, Computer Network, Medical, Military & Aerospace etc.

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2.1 Radio Waves

- The type of electromagnetic waves which are transmitted in every direction of free space.
- It is used in Wide Area Network or Larger geographical areas.
- This wave has a 3 KHz to 1 GHz frequency range.
- It provides a higher data transmission rate.
- The sending and receiving antenna are not aligned.
- It is useful for multicasting when there is one sender and many receivers.

➤ Applications:

1. FM Radio
2. Mobiles
3. Television.

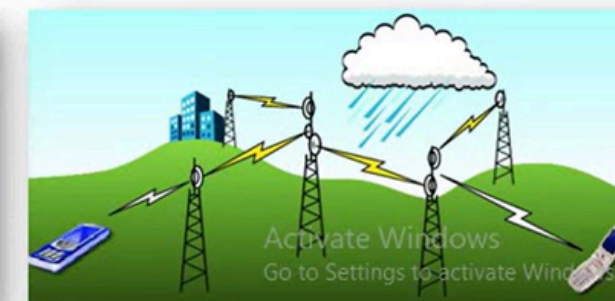
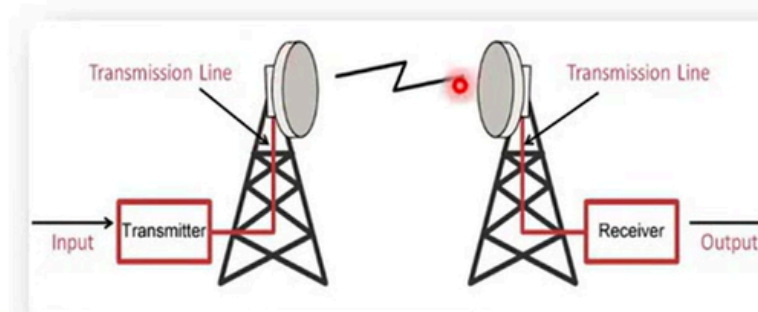


2.2 Macro Waves

- It uses antennas as the main element for sending and receiving the data.
- The distance covered by the signal is directly proportional to the height of the antenna.
- The signal range for this type of transmission is between 1 GHz and 300 GHz.
- They carry large amount information with capacity.
- Satellite microwaves can communicate via any device globally.

➤ Applications:

1. Mobiles
2. Television networks.
3. Satellite networks.



2.3 Infrared

- It is a wireless technology used for communication over extremely short distance.
- The frequency range is from 300 GHz to 400 THz.
- Data cannot pass through the obstacles.
- It provides high bandwidth & security.
- It holds lower information rate of transmission.

➤ Applications:

1. TV Remote
2. Wireless Mouse, Keyboard, Printer.
3. Transfer data between two Mobiles.

